Algorithmic & AI Promo Discovery for Vertical Microdramas

Candidate: Saurabh Solanki | Live Prototype: https://kuku-promo-finder-saurabh105.streamlit.app/

GitHub Repository: https://github.com/100rabhsolanki/kuku-promo-finder | Dataset: 50 Episodes (3,727 blocks)

1. PROBLEM CHOICE, TARGET USER & EDITORIAL WORKFLOW

Why Problem 1: In high-velocity microdrama publishing, user acquisition depends entirely on short-form promo ads (20-60s) on TikTok, Instagram Reels, and YouTube Shorts. The bottleneck is not rendering, but creative discovery: human editors spend 1.5 to 2 hours manually scrubbing through 50 episodes per series to find hooks, reveals, and cliffhangers.

Target User & Transformation: Microdrama Video Editors and Growth Marketers. The tool serves as an algorithmic scout, transforming a 100-minute manual video scrub into a 3-minute review of pre-scored, timestamped candidate promo cards.

2. OPERATIONAL ASSUMPTIONS & TIMING REALITIES

Screenplay Timestamps vs. Finished Video Runtime: Timestamps represent planned script timings. Editorial pauses or bumpers require editor verification. The tool anchors cuts to verified script timestamps rather than estimating untracked pacing.

Multi-Track Concurrency: CAMERA, ACTION, and DIALOGUE overlap in the screenplay. Clip bounds are computed from the earliest starting event to the latest ending event of the dramatic beat without duplicate duration accumulation.

Scroll-Stopping Hook Window: Social video drop-off occurs within 2.5-3.0s. Hooks occurring after 3.5s are heavily penalized.

Single vs. Multi-Episode Promos: Continuous single-episode clips offer optimal narrative cohesion; multi-segment promos are curated for high-contrast pairs (e.g., Altar Betrayal in Ep 1 paired with Underdog Marriage in Ep 4).

3. SYSTEM ARCHITECTURE & RESPONSIBILITY DIVISION

We enforce a strict separation between deterministic code/rules and heuristic/AI scoring:

HANDLED BY DETERMINISTIC CODE / RULES	HANDLED BY SCORING ENGINE / AI MODEL
- Script block regex parsing (3,727 blocks parsed, 0 missed) - Strict 20-60s window math and timestamp arithmetic - Non-Maximum Suppression (NMS) IoU overlap deduplication	- Hook immediacy, shock factor & verbal jolt evaluation - Cliffhanger urgency & unresolved dramatic stakes - Standalone clarity & context-dependency penalty

4. MULTI-CRITERIA SCORING & RANKING FORMULATION

Composite Score = 0.30(Hook) + 0.25(Cliffhanger) + 0.20(Standalone) + 0.15(Intensity) + 0.10(Context)

1. Hook Immediacy (0-10): Scored by linguistic shock (e.g. 'You're the Duke?', 'Married a beggar!'), physical aggression (slap, blade, shattered gate), and sound stingers within the first 3.5s. Decays by 40%/s for every second of delay.
2. Cliffhanger Urgency (0-10): Unanswered questions, cut-to-black scene peaks, interrupted revelations, and red-hot iron threats.
3. Standalone Clarity (0-10): Rewards universal archetypes (betrayal at altar, undercover Duke); penalizes niche medical/lore jargon.
4. Temporal Deduplication: Retains top-scoring moments while eliminating 6,377 overlapping time-shifted candidate windows.

KUKU AI GENERALIST ASSIGNMENT | PROBLEM 1: QUALITY AUDIT & PRODUCTION SCALING

5. TOP-RANKED PROMO CANDIDATES (PROVEN FROM 50-EPISODE RUN)

Rank 1 (Ep 45, 54.6s | Score 9.02): Undercover Duke Reveal. Hook: Juliana discovers 'You're the Duke?' at 00:22.90. Climax: Alexander frees Juliana from chains; Catherine screams 'Wife?!' in public horror at 01:10.50.

Rank 2 (Ep 42, 57.8s | Score 8.84): Shattered Cathedral Gates. Hook: 'Guards! Arrest this imposter!' at 00:18.00. Climax: Alexander emerges in full ducal regalia; Duchess reveals: 'Let everyone hear me! This is my son Alexander!'

Rank 3 (Ep 11, 57.1s | Score 8.15): Class Betrayal Confrontation. Hook: Gerald mocks 'You married a beggar!' at 00:29.30. Climax: Gerald dares him: 'Going to strike a gentleman in front of his wife?' at 01:24.00 before physical confrontation.

Rank 4 (Ep 11 + Ep 42, 36.3s | Score 8.10): Mockery to Ducal Vindication Multi-Segment. Gerald mocks Juliana's beggar husband, cross-cut to Ep 42 where Alexander's mother proclaims him the rightful Duke before the royal guards.

6. HONEST QUALITY CHECK: REAL FAILURE MODES & EDGE CASES

Automated systems fail when narrative context cannot be deduced from isolated words. We observed three failures:

Failure 1 (False Positive - Medical Cues in Ep 8 & 9): Words like 'blood', 'fever', and 'wound' scored high on intensity but represented quiet herbal caretaking rather than dynamic viral conflict. Fix: Added penalties for non-combative medical scenes.

Failure 2 (Context-Trapped Climax in Ep 29): High dungeon tension scored well, but dialogue centered on 'charcoal antidotes' and 'mother's grimoire'. A cold TikTok scroller has zero idea who Rupert is. Fix: Standalone comprehensibility penalizes lore-density.

Failure 3 (Establishing Shot Lag in Ep 7): Insult occurs at 00:08.50 after silent door opening. If clipped from 00:00.00, the promo loses the viewer. Fix: Window start dynamically snaps to the exact timestamp of the hook beat rather than scene start.

7. HUMAN-IN-THE-LOOP EDITORIAL PROTOCOL

The tool proposes candidate cards; the human video editor confirms final cuts via a 3-step checklist:

1. Visual Jolt Audit: Verify the opening 0-2s has immediate motion or punch-in (front-trim if actor hesitation occurs).
2. Audio Tail Audit: Confirm dialogue does not cut mid-sentence; verify music swell resolves into an unresolved blackout.
3. Channel Destination: Route betrayal/slap clips (Ep 1, 11) to TikTok; route vindication/identity reveals (Ep 42, 45) to YouTube Shorts.

8. AI TOOLS USED & WHERE THE MODEL GOT SOMETHING WRONG

AI Tools: Gemini 3.8 Flash / Claude 3.5 Sonnet for archetype analysis, rubric calibration, and failure-mode clustering.

Where the AI Got It Wrong: When analyzing Ep 1, an LLM flagged Juliana's tearful clenching of her bridal gown (02:11.30) as a 'resolved resolution' because the groom had finished speaking and walked away. The AI mistook the physical end of an argument for emotional closure, missing that Juliana's hardening glare was the ultimate microdrama cliffhanger setup.

9. SCALING ACROSS SHOWS & PRODUCTION TIMELINE

Production Scaling: The Python engine processed 50 episodes (102.6 minutes of drama, 3,727 blocks) in 2.01 seconds.

Cost at Scale: $0.00 for rule-based parsing and NMS deduplication. When connected to a lightweight LLM API (e.g., Gemini Flash), evaluating the 144 candidate windows costs less than $0.01 per show, compared to $40+ of human editor scrubbing time.